

## Amber Fredenburg

Personal website: <https://amberfredenburg.github.io/>

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### Academic History

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| <b>PhD</b> | The Ohio State University, Columbus, Ohio<br>Evolution, Ecology, and Organismal Biology<br>Dissertation: <i>Digging deeper: Unearthing the nesting requirements of ground-nesting bees</i><br>Graduate Minor, Public Policy | 2026 |
| <b>MS</b>  | Lakehead University, Thunder Bay, Ontario<br>Forestry<br>Thesis: <i>Diversity, phenology, and host associations of wild bees (Hymenoptera: Anthophila) in Thunder Bay, Ontario</i>                                          | 2020 |
| <b>BS</b>  | The Ohio State University, Columbus, Ohio<br>Zoology                                                                                                                                                                        | 2018 |

### Professional Experience

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| Research Associate, The Ohio State University<br>Department of Evolution, Ecology, and Organismal Biology<br><i>The Ohio Bee Survey and Endangered Pollinator eDNA Project</i>          | 2026–Present |
| Graduate Research Associate, The Ohio State University<br>Department of Evolution, Ecology, and Organismal Biology<br><i>The Ohio Bee Survey and Specialist Bee Project</i>             | 2022–2026    |
| Graduate Teaching Assistant, Ohio State University<br>Department of Evolution, Ecology, and Organismal Biology<br><i>Ecology and Organismal Diversity</i>                               | 2021–2024    |
| Graduate Teaching Assistant, Ohio State University<br>Center for Life Sciences Education<br><i>Biological Sciences: Form, Function, Diversity, and Ecology</i>                          | 2021         |
| Graduate Teaching Assistant, Lakehead University<br>Department of Natural Resources Management<br><i>Forest Entomology and Population Ecology</i>                                       | 2019–2020    |
| Undergraduate Research Assistant, The Ohio State University<br>Department of Evolution, Ecology, and Organismal Biology<br><i>Pumpkin Pollinator Project and Ohio Bumble Bee Survey</i> | 2017–2018    |

### Publications

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- Fredenburg, A. N., J. Gibbs, and D. C. Henne. (2021). A survey of wild bees (Hymenoptera: Anthophila) and their floral associations in Thunder Bay, Ontario. *Journal of the Kansas Entomological Society*. 93(3): 211-229. <https://doi.org/10.2317/0022-8567-93.3.211>
- Fredenburg, A. and K. Goodell. (2025). Soil sanctuaries: Experimental manipulations enhance ground-nesting bee habitat across an urban gradient. *bioRxiv*. <https://doi.org/10.1101/2025.04.18.649528R>
- Pen, I. A. M., Dueñas, L. F., Armondo Sánchez, J., Vergera-Flórez, D. C., Gómez, C. E., Tovar-Bolaños, V., Palacios, J. A., Moreno, J. A., Stoilova, M. I., Lo, C., Kephart, C., Fredenburg, A., Muñoz, R. G., Simões, N., Lizaloe-Gullermo, J. & Daly, M. (2026). New reports of *Bunodeopsis Andres*, 1881 in the Greater Antilles, Southern, Southeastern, and Western Caribbean. *Marine Ecology*. 47(2): e70081. <https://doi.org/10.1111/maec.70081>

- Fredenburg, A., and K. Goodell. The Role of Nesting Resources on Specialist Bee Populations and the Bee Community (in prep).
- Goodell, K., Fredenburg, A., Spring, M., and Semler, N. The bees of Ohio: using statewide survey data to strategize future monitoring (in prep).
- Fredenburg, A., and K. Goodell. Soil characteristics and glacial history structure ground-nesting bee communities across Ohio (in prep).
- Fredenburg, A. (2026). Digging deeper: Unearthing the nesting requirements of ground-nesting bees. *ProQuest* (in press).

## **Presentations and Posters**

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- 2026** Fredenburg, A – Digging deeper: Unearthing the resource requirements of ground-nesting bees, *Evolution, Ecology, and Organismal Biology Departmental Seminar*, Ohio State University, Columbus, Ohio.
- 2025** Fredenburg, A., Goodell, K., Spring, M., & Semler, N.—Capturing Ohio’s bee diversity by surveying the host plants of diet specialists, *Oak Openings GRI Science Summit*, Toledo, Ohio
- Fredenburg, A. & Goodell, K.—Bee responses to resources depend on ecological scale, *Ecological Society of America*, Baltimore, Maryland
- 2024** Fredenburg, A., & Goodell, K.—Exploring the link between soil characteristics and the bee across Ohio, *Natural Resource Research and Management Symposium*, Cleveland, Ohio
- Goodell, K., Spring, M., Fredenburg, A., Semler, N., & Feters Fogarty, A.—The Ohio Bee Survey: species checklist and planning future monitoring, *Natural Resource Research and Management Symposium*, Cleveland, Ohio
- Fredenburg, A. & Goodell, K.—Enhancing the nesting habitat for ground-nesting bees, invited symposium speaker, *Entomological Society of America*, Pheonix, Arizona
- Goodell, K., Fredenburg, A., Semler, N., & Spring, M.—Optimizing wild bee surveys for conservation management: a comparison of survey methods, *Entomological Society of America*, Pheonix, Arizona
- 2023** Fredenburg, A., Spring, M., & Goodell, K.—Documenting specialist bee diversity and factors that influence their distributions in Ohio, *Ohio Natural History Conference*, Cincinnati, Ohio
- Semler, N., Spring, M., Fredenburg, A., & Goodell, K.—Native and non-native thistle (*Cirsium* spp.) support many generalists, but fewer specialist bees, *Ohio Invasive Plant Research Conference*, Columbus, Ohio
- Fredenburg, A., Spring, M., & Goodell, K.—The role of nesting resources on bee abundance at host plant patches, *Entomological Society of America*, National Harbor, Maryland
- Goodell, K., Spring, M., Helton, C., & Fredenburg, A., Semler, N., & Spring, M.—Lessons from the Ohio Bee Survey: Analysis of pan traps and suggestions for improving specimen richness while limiting specimen numbers, *Entomological Society of America*, National Harbor, Maryland
- 2022** Goodell, K., Spring, M., Helton, C., & Fredenburg, A.—The role of midwestern USA forests in Supporting bee populations, *Entomological Society of America*, Vancouver, British Columbia
- 2020** Fredenburg, A., & Henne, D.—Native bee composition and diversity at three sites in Thunder Bay, Ontario, *Lakehead University Graduate Studies Thesis Presentations*, Thunder Bay, Ontario
- Fredenburg, A., & Henne, D.—Biodiversity of native bees at three sites in Thunder Bay, Ontario, *Graduate Studies Research Conference Poster Competition* (Winner), Thunder Bay, Ontario

## **Undergraduates Mentored**

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### **Ohio State University**

Rex Harvey-Thurston, *Ground-Nesting Plot Project* and *Specialist Bee Project* (2023-present)

Abby Subler, *Specialist Bee Project* (2024-2025)

Noma Semler, *Specialist Bee Nesting Project* (2022-2023)

Vanessa Chilcoat, *Diversity and Distribution of Asilidae of Ohio* (2022)

**Lakehead University**

Alena Albertson-Kok, *Diversity of Saproxyllic Beetles in Thunder Bay, ON* (2019-2020)

## **Community Engagement**

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Convening Conservationists Talk, Great Parks of Hamilton County, Cincinnati, Ohio (4/21/2026) – disseminated Ohio Bee Survey results and discussed guidelines for developing a bee monitoring program and how to construct effective pollinator habitat.

Science Olympiad Entomology B, (October 2024-April 2025)—wrote the Entomology exam with a team of co-authors from Ohio State University for middle school and high school students participating in Science Olympiad 2025.

Franklin County Pollinator Pathways, Franklin Soil and Water Conservation, Columbus, Ohio (6/6/2024)—invited as a guest bee expert for a gathering of representatives, conservation groups, advocacy groups, and other organizations to discuss how to create more and less fragmented native pollinator habitat in Franklin County.

FLOW Webinar: Creating habitat for our native ground-nesting bees, Friends of the Lower Olentangy Watershed, Columbus, Ohio (5/20/2024)—lead a webinar for members and anyone interested in creating ground-nesting bee habitat.

FLOW Science Board Meeting, Friends of the Lower Olentangy Watershed, Columbus, Ohio (1/8/2024)—discuss research results with the science committee of FLOW and discuss ways to improve pollinator gardens for nesting bees.

Native Bees and Where They Nest, Beecoming Home, The Bee Collective, Columbus, Ohio (9/9/2023)—engaged the public with information about native bee nesting at a collaborative art and science event.

Science Day at Parkmoor Elementary School, Scientific Thinkers, Ohio State University (12/8/2022), —lead volunteer instructor for a strawberry DNA removal experiment in 3<sup>rd</sup> grade classrooms.

Bees, I Love Them, and You Should Too—presentation to engage K-12 students in the importance of bee conservation, Driscoll Elementary, Centerville City Schools, Centerville, Ohio, (4/15/2020) and Fredrick Douglas Elementary, Cincinnati City Schools, Cincinnati, Ohio (5/4/2023)

## **Professional Service**

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Journal of Insect Behavior – Reviewer (2025, 1 manuscript)

Journal of Insect Conservation – Reviewer (2025, 1 manuscript)

Journal of Applied Ecology – Reviewer (2024, 2 manuscripts)

Journal of Restoration Ecology – Reviewer (2022-2023, 2 manuscripts)

Ecology and Evolution – Reviewer (2022, 1 manuscript)

## **Grants Received**

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2021 ENGIE-Axiom Research Incentive Recruitment Scholarship Award. \$5,000.

2023 Garden Club of America Pollinator Fellowship. \$4,000.

## **References**

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Karen Goodell, Ph.D.,

Professor, Ohio State University, Evolution, Ecology, and Organismal Biology

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Frances Sivakoff, Ph.D.

Professor, Ohio State University, Evolution, Ecology, and Organismal Biology

Contact: [Sivakoff.3@osu.edu](mailto:Sivakoff.3@osu.edu)

Steve Hovick, Ph.D.

Professor, Ohio State University, Evolution, Ecology, and Organismal Biology

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